Artificial General Intelligence in Healthcare

A case for AGI as a CrossSpeciality Diagnostic Reasoner

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Course: ITAI 2732

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Date: May 3rd 2026

Introduction

Healthcare is one of the most data-rich and decision-dense industries in the modern economy, yet it remains one of the slowest to translate raw information into reliable judgement at the individual patient level. A patient walking into a primary-care clinic generates an enormous trail of evidence. Evidence like vitals, imaging, lab values, genomic markers, prior history, and social determinants. All of which a clinician must compress into a diagnosis and treatment plan in roughly fifteen minutes. The cost of getting that synthesis wrong is devastating: a 2023 Johns Hopkins analysis estimated that diagnostic errors are responsible for approximately 795,000 deaths and permanent disabilities in the United States each year, with vascular events, infections, and cancers (the "Big Three") accounting for 75% of those serious harms (Newman-Toker et al., 2023).

Artificial intelligence (AI) has been promoted for over a decade as a solution to this diagnostic burden, and the regulatory record reflects substantial progress: by the end of 2025, the U.S. Food and Drug Administration (FDA) had authorized over 1,400 AI-enabled medical devices, with 295 cleared in 2025 alone (Innolitics, 2025; IntuitionLabs, 2026). However, this entire regulatory portfolio is composed of what researchers now call

Industry Analysis: The Current State of Diagnostic AI

A Specialty-Locked Landscape

The most striking feature of medical AI today is its concentration. Of the more than 1,300 AI-enabled medical devices approved by the FDA, approximately 77% are radiology applications (Sivakumar et al., 2025; The Imaging Wire, 2025). Cardiovascular tools account for another 10%, and neurology, hematology, ophthalmology, and other specialties together make up the remaining 13%. This distribution is not a coincidence; it reflects how narrow AI is built. Each system is trained on a single, well-bounded task, like detecting lung nodules on a chest X-ray, segmenting a tumour on an MRI, or classifying a retinal photograph, and is incapable of operating outside the boundary it was trained on. The result is a regulatory landscape that looks less like a coherent diagnostic platform and more like a thousand single-purpose appliances stacked on top of one another (see Figure 1).

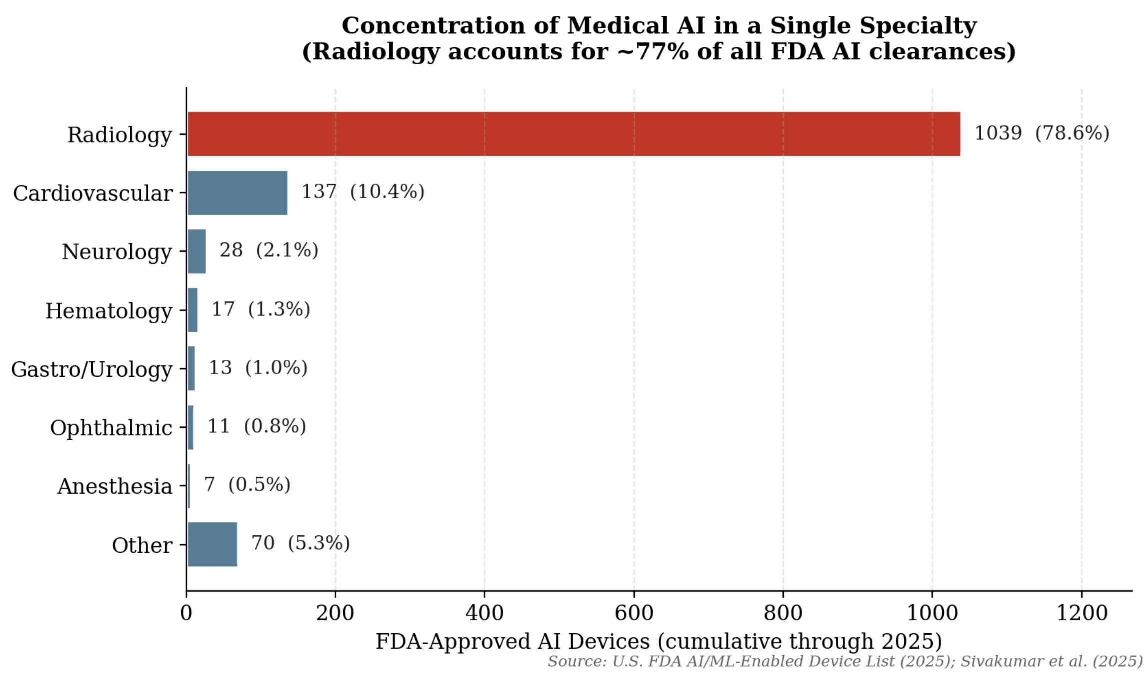

Figure 1. Distribution of FDA-approved AI medical devices by specialty (cumulative through 2025). Radiology dominates the regulatory landscape, illustrating the specialty-locked nature of current narrow AI.

Three Limitations of Narrow AI in Diagnosis

Three limitations of the current approach are particularly relevant to the AGI case. First, narrow models cannot transfer knowledge across domains. A chest X-ray classifier cannot read a CT scan; a CT model cannot read a histopathology slide; none of these can integrate the patient's medication history. This forces clinicians to act as the integration layer, mentally combining outputs from multiple disconnected systems. This results in a workflow that re-introduces the very cognitive load AI was supposed to relieve.

Second, narrow models lack robust generalisation. A systematic review of FDA-cleared radiology AI found that only 5% of devices underwent prospective testing and only 8% included a human-in-the-loop evaluation; the vast majority were cleared via the 510(k) pathway by demonstrating equivalence to a previously approved device rather than through new clinical evidence (Sivakumar et al., 2025). Models trained on data from one hospital system frequently degrade when deployed at another, where imaging equipment, patient demographics, and disease prevalence differ.

Third, narrow AI cannot reason about novel situations. Diagnostic medicine is full of cases that fall outside textbook patterns: a stroke that mimics migraine, a rare metabolic disorder that presents as psychiatric symptoms, an autoimmune disease that masquerades as cancer. These are precisely the cases that drive the Big Three diagnostic-error figures, and they are precisely the cases that current narrow AI, which has no concept of what lies outside its training distribution, is least equipped to help with (Newman-Toker et al., 2023).

AGI Application Proposal: The Cross-Specialty Diagnostic Reasoner

Defining AGI

Artificial general intelligence is widely defined in current literature as a hypothetical AI system capable of *autonomous learning, generalisation across domains, and reasoning at or above human cognitive capability across an open-ended range of tasks* (Armitage, 2025; Google Cloud, 2025). The distinguishing features that separate AGI from narrow AI can be summarised in four points:

- **Cross-domain transfer.** Knowledge acquired in one area (e.g., immunology) can be applied to a different area (e.g., dermatology) without retraining.
- **Open-ended adaptation.** The system can confront problems it was never explicitly trained on and reason its way to a solution.
- **Compositional reasoning.** Multiple types of evidence, such as imaging, text, lab values, genomics, and lifestyle, can be combined into a single coherent inference.
- **Autonomous goal pursuit.** The system can plan multi-step actions (e.g., ordering an additional test, comparing to similar prior cases) rather than simply returning a single classification.

No system available today meets all four criteria. Multimodal foundation models such as GPT-class systems and Google's Gemini are sometimes described as being on the path toward AGI, but they remain narrow in the sense that they are not deployed with full autonomy in clinical settings and cannot reliably perform cross-domain medical reasoning at expert level (Armitage, 2025). The proposal advanced here is not that AGI is available now, but that healthcare is the industry where its eventual arrival would matter most, and where serious thinking about deployment should begin in advance of the technology being mature.

The Proposed Application

The application is a hospital-wide AGI system functioning as a cross-specialty diagnostic reasoner. When a patient presents with an undifferentiated complaint, the system would ingest the entirety of available evidence. Such as imaging from any modality, laboratory results, genomic data, the patient's electronic health record, current medication list, and family history. And with this evidence, the system produces a ranked differential diagnosis along with an explicit reasoning trace and a recommended next-step diagnostic plan. Crucially, the same system would handle a chest X-ray that suggests pneumonia, an MRI that suggests multiple sclerosis, a lab panel that suggests sepsis, and a constellation of vague symptoms that suggests a rare metabolic disease, *all without retraining and within a single conversation with the clinician*. Figure 2 contrasts this workflow with the current narrow-AI alternative.

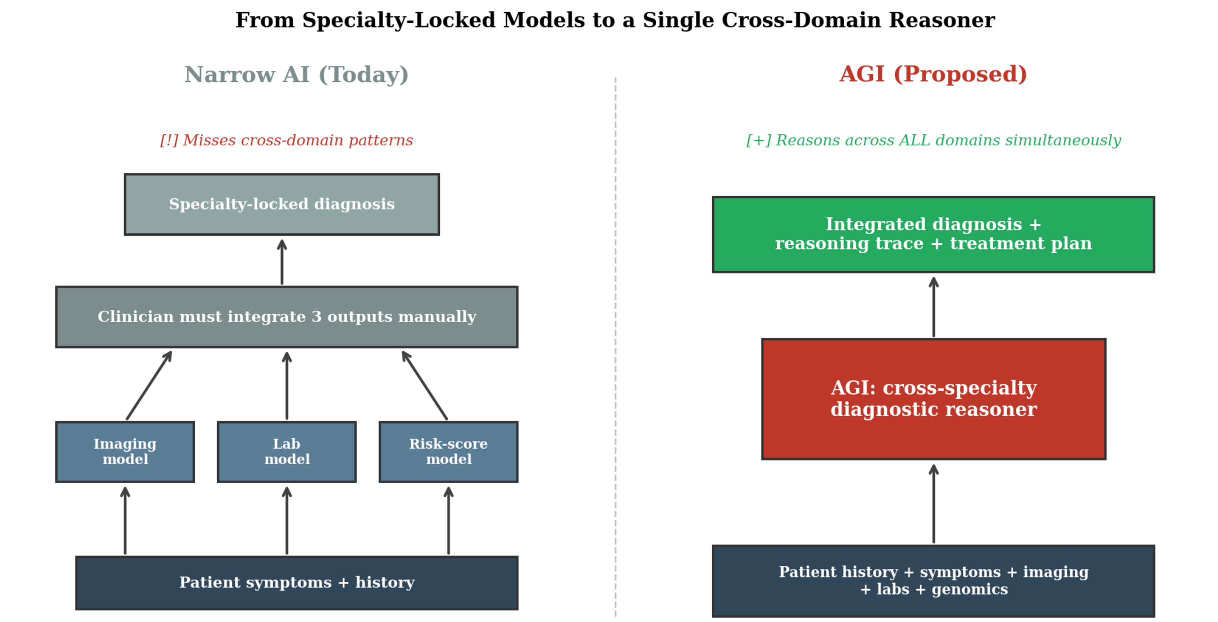

Figure 2. Narrow AI today (left) requires the clinician to manually integrate outputs from multiple specialty-locked models. AGI (right) would replace this fragmented pipeline with a single cross-domain reasoner.

Anticipated Benefits

The benefits would fall into three categories. **Diagnostic accuracy and lives saved.** If even a fraction of the 795,000 annual misdiagnosis-related deaths and permanent disabilities in the United States are attributable to failures of cross-domain reasoning rather than failures of within-domain perception, an AGI system that genuinely integrates evidence across specialties could prevent tens of thousands of these harms each year (Newman-Toker et al., 2023). **Equity and accessibility.** A specialist-level diagnostic reasoner deployable as software could, in principle, bring tertiary-care-quality reasoning to community hospitals, rural clinics, and low- and middle-income countries that currently lack access to subspecialty expertise altogether (Just Think AI, 2024). **Workflow integration.** Rather than the current model in which clinicians manually combine outputs from a dozen disconnected narrow tools, an AGI system would produce a single integrated assessment with an auditable reasoning trace, reducing cognitive load and freeing clinicians for the human-judgement and patient-relationship work that remains uniquely theirs.

Risks and Ethical Concerns

The risk profile of AGI in healthcare is substantially more demanding than that of narrow AI, and three categories deserve particular attention.

Bias and equity failure modes.

Medical AI inherits the biases of its training data, and the data available for training tends to be drawn from well-resourced, predominantly Western, predominantly white populations. An AGI system that confidently reasons across all of medicine could amplify these biases at scale rather than confine them to a single specialty (Salahuddin et al., 2025). Mitigation requires deliberate inclusion of underrepresented populations in training data, ongoing post-deployment fairness audits, and transparency about the populations on which the system performs poorly.

Privacy and data governance.

A system that ingests a patient's entire health record, including genomic data, mental-health notes, and pharmacy history, creates an unprecedented attack surface. Strong encryption, federated training architectures that keep data on local servers, granular consent mechanisms, and clearly enforced retention limits are necessary preconditions for deployment.

Workforce and de-skilling effects.

If AGI handles the integration step that currently distinguishes the best clinicians, there is a real risk that clinical reasoning skills atrophy in junior physicians who train alongside the system. This is not a reason to refuse deployment, but it is a reason to design AGI systems as collaborators that explain their reasoning rather than as oracles that hand down conclusions, so that clinicians remain genuine participants in the diagnostic process (Armitage, 2025).

Existential and dual-use risks.

Recent academic literature has begun to take seriously the possibility that AGI itself, not merely its misuse, could pose risks to public health, including risks of misalignment between system objectives and human values, and dual-use applications such as the design of pathogens or the optimisation of disinformation campaigns (Armitage, 2025). Healthcare deployment must therefore be coupled with broader AI-safety research, not treated as a domain-specific problem.

Conclusion

Healthcare is the industry where the gap between current narrow AI and a true AGI system is most consequential. The 1,400-plus AI medical devices currently authorized by the FDA represent real progress, but their concentration in a single specialty, radiology, reveals the underlying truth: today's medical AI is a collection of single-purpose tools, not a diagnostic reasoner. The 795,000 annual deaths and permanent disabilities attributable to diagnostic error in the United States are largely caused by failures of integration across domains, exactly the kind of problem that narrow AI is structurally incapable of addressing.

As a system capable of cross-domain transfer, open-ended adaptation, compositional reasoning, and autonomous goal pursuit, AGI represents a categorically different proposition. Deployed thoughtfully as a cross-specialty diagnostic reasoner, AGI could prevent a meaningful share of those harms, extend specialist-level reasoning to populations currently lacking access, and integrate naturally with clinical workflows in a way that today's fragmented tools cannot.

The path to that future is not, however, a path of pure optimism. The risks of amplified bias, privacy exposure, de-skilling, and the broader category of AGI-safety concerns now appearing in the public-health literature are real and serious. The thoughtful response is neither to dismiss AGI as science fiction nor to deploy it prematurely, but to begin the regulatory, ethical, and clinical-evidence work now, while the technology is still maturing. Healthcare has the most to gain from AGI and the most to lose if it is deployed badly. Both facts argue for the same conclusion: serious thinking about AGI in medicine should not wait until AGI arrives.

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